3.3.5 Wrap-Up: Error Analysis

The table of values represents a linear relationship.

x	y
1	-17.3
6	-13.8
10	-11
15	-7.5

Kayla and George wrote the equation of the line representing the linear relationship. They both began by finding the slope, m, and then writing the equation of the line in point-slope form. Both students made errors.

Kayla's Work	George's Work
$m = \frac{-11-7.5}{10-15} = \frac{-18.5}{-5} = 3.7$ $y + 11 = 3.7(x - 10)$	$m = \frac{-13.8-(-11)}{6-10} = \frac{-2.8}{-4} = 0.7$ $y - 11 = 0.7(x - 10)$

1. Circle each student's error.

2. Write a short note to each student to explain their mistake.

To Kayla:	To George:

Unit 3, Lesson 4: Writing Equations of Lines From Written Descriptions

Benchmarks

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description, or a table of values within a mathematical or real-world context.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

Learning Target

☐ I can write the equation of a line given a written description.

3.4.1 Warm-Up: A Day at Work for a Personal Trainer

Tiffany Ann Fields is a health coach and personal trainer for UC Berkeley's WorkStrong program and Studio 360. Tiffany followed a path from track and field in high school, to college, into her current career.

1. Tiffany said that as a personal trainer she always starts with her client's goals and builds the program from there. Why do you think she recommends this as a best practice?

2. In an interview about her career, Tiffany said, "I think the best thing you can do in life is be comfortable feeling like you're not good at something and just learning from it." Explain whether or not you agree with this statement and why.

3.4.2 Collaboration: Finding the Slope and y-intercept of a Line From Equations

Work with your partner to complete the following.

1. The equations of four lines are given in the table.

   a. For each equation, identify the slope and y-intercept of the line. Use the space provided below the table to rewrite the equations in different forms, as needed, to complete the table.

Equation	Slope	y-intercept
$f(x) = \frac{x}{2} + 1$		
$y - 5 = 6(x - 4)$		
$\frac{1}{3}(9x + 12) = y$		
$-3x + 2y = -8$		

   b. Which form of the equation of a line is easiest to find the slope and y-intercept of the line?

3.4.3 Guided Instruction: Writing Equations of Lines From Descriptions

1. A line passes through the points $(-1, 10.75)$ and $(4, 7)$.

   a. What feature(s) of the line can be determined from this information?

   b. Write the equation of the line in point-slope form.

   c. Rewrite the equation in standard form.

2. Julian measured the amount of rain that had fallen using a rain barrel. The rain fell at a constant rate from 12:00 p.m. to 4:00 p.m. He began recording at noon when the height of the rain in the barrel was 1.4 inches. By 4:00 p.m., there were 2.84 inches of rain in the barrel.

   a. Determine two ordered pairs on the graph of this relationship.

   b. Use the ordered pairs to write the equation of the line in slope-intercept form.

   c. Use the same information to write the equation of the line in point-slope form.

An ***intercept*** is the value of a variable when all other variables in the equation equal 0. On a graph, the intercept is the value where a function crosses an axis.

The ***x-intercept*** is the value of x at the point where a line or graph intersects the x-axis. The value of y is 0 at this point.

3. The graph of a linear relationship has an x-intercept of 4 and a y-intercept of -3.2.

a. Write the ordered pairs to indicate the location of the intercepts on the graph of the line.

Key Feature of the Line	Location
x-intercept	
y-intercept	

b. Write the equation of the line with these intercepts in point-slope form.

The equation of any line can be determined given two points, a point and the slope, or the intercepts of the line. Either slope-intercept form or point-slope form can be written from the given information, and either of these forms can be used to rewrite the equation in standard form.

3.4.4 Guided Practice: Writing Equations of Lines From Descriptions

1. A line has a slope of $\frac{9}{11}$ and a y-intercept at $(0, -14)$.

a. Write the equation of the line in standard form.

b. In which form of the line did you first write the equation in order to find the standard form?

c. Explain why you selected that form based on the information given.

2. A line passes through the points $(18, -1.9)$ and $(30, 12.5)$.

a. Write the equation of the line in any form you choose.

b. In which form of the line did you write the equation?

c. Explain why you selected that form based on the information given.

3.4.5 Collaboration: Linear Equations

Work with your partner to complete the following.

1. A line has a slope of $\frac{3}{5}$ and passes through the point $(-6, 20.4)$.

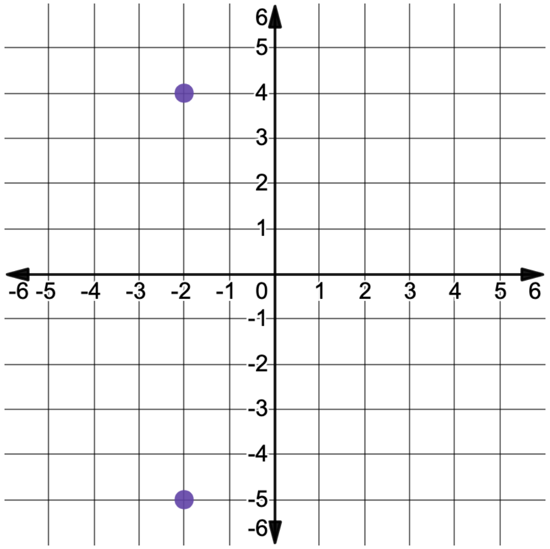
a. Write the equation of the line in point-slope form.

b. Discuss with your partner how to use the equation from Part A to find the intercepts of the line. Write a brief summary of your discussion.

c. What is the x-intercept of the line?

d. What is the y-intercept of the line?

2. The points $(-2,-5)$ and $(-2,4)$ are plotted on the coordinate grid shown.



a. A line passes through these points. Write the equation of the line.

b. Complete the statements.

The slope of the line is
○ 0.
○ undefined.
The
○ x-intercept
○ y-intercept
of the line is -2, and the line does not have a(n)
○ x-intercept.
○ y-intercept.

3. The equation of a line is $4x - 28y = 70$.

a. Discuss with your partner which form of the equation of the line most clearly the reveals the value of y when $x = 0$.

b. Rewrite the equation in this form.

c. What is the value of y when $x = 0$?

d. Describe how to find this point on the graph of the line.

3.4.6 Wrap-Up: Ticket Out the Door

1. A line passes through the points $(-2.7, 0.78)$ and $(21, 15)$. Write the equation of the line in standard form.

Unit 3, Lesson 5: Scatter Plots and Linear Relationships

Benchmark

MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

Learning Targets

- ☐ I can informally fit a straight line to model the relationship between two variables in a scatter plot.
- ☐ I can interpret the slope and y-intercept of a linear function fit to bivariate numerical data in context.
- ☐ I can determine an appropriate domain and range for bivariate numerical data based on the context.